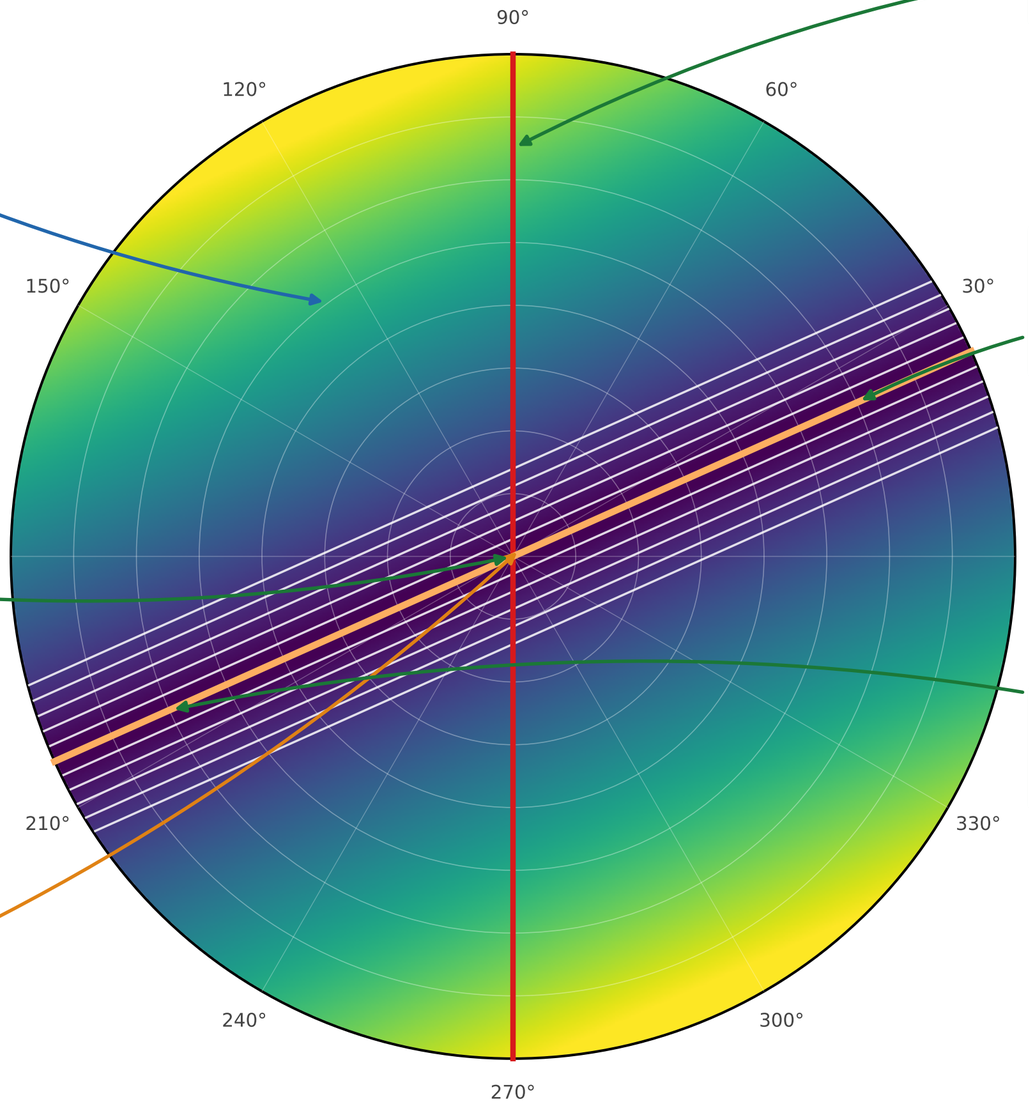
Stage 3 — The pure wedge: a² = 4q (double line)

deg = (m − (a/2)·n)² with a = 2√5: rank 1, the conic degenerates to a doubled line. Colour = √deg = |m − √5·n| — constant along lines parallel to it.

nested rings = weight strata |z| = q^(−σ)
= attenuation classes; purity = one ring
= distortionless (÷ q^½ → lossless)
[Restatement for Frobenius / Analogy for ζ]

y-axis = critical line Re s = ½:
the unitary spectrum of scaling
U_λ f(x) = λ^½ f(λx) on half-densities
(|λ̄^(½−s)| = 1 ∀λ ⇔ Re s = ½; TAA Lemma 153 §5)
[Theorem; frame axis = Analogy]



double line a² = 4q: deg = (m − (a/2)n)²;
Frobenius = ±√q·I (not unipotent); F = V
angle here = atan(1/√5) ≈ 24°, set by a/2 —
not 45°
[Theorem]

dual of a line pair = its vertex counted twice
(adjugate rank 1); Morse: the critical level
of a saddle is a crossing pair
[Theorem]

wedge = null cone: on the hyperbolic side
the level deg = 0 opens into two lines =
characteristics x ± ct of dx² − c²dt²
[Theorem]

origin ↔ point at infinity = the cusp:
constant term $a_0(y) = y^s + \varphi(s)\cdot y^{(1-s)}$,
$|\varphi(\tfrac{1}{2}+it)| = 1$ (TAA Lemma 153 NUM ≤ 2e−31)
[Analogy for the identification]

Status tags: [Theorem] [Restatement] [Analogy] [Numerical] [Cited] — see legend.png. RH is not proved. No null rays at 45° (per J. W. McGreevy); lobes are elliptical conic sections, not elliptic curves.

After J. W. McGreevy's animation 'Flat Potential Oscillations' (Aug 2025, x.com/McGreevyMD/status/1951474097990279331). Re-rendered by Grok Bot from the analysed geometry; not J. W. McGreevy's code.